

MA.912.GR.1.6

Benchmark

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

Benchmark Clarifications:

Clarification 1: Instruction includes demonstrating that two-dimensional figures are congruent or similar based on given information.

Connecting Benchmarks/Horizontal Alignment

- MA.912.GR.2
- MA.912.GR.3.3
- MA.912.GR.4.3
- MA.912.GR.6.5

Terms from the K-12 Glossary

- Congruent
- Similarity

Vertical Alignment

Previous Benchmarks

- MA.8.GR.2.4
- MA.912.AR.2.1

Next Benchmarks

- MA.912.T.2

Purpose and Instructional Strategies

In grade 8 students solved real-world and mathematical problems involving similar triangles, defining similarity in terms of dilations. In Geometry, students extend previous understanding of congruence and similarity to solve mathematical and real-world problems involving

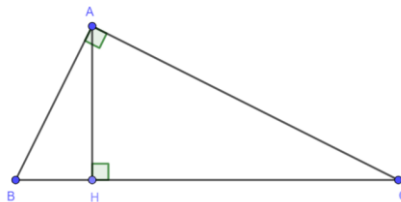
congruent or similar polygons. In later courses, students will use similar triangles to develop trigonometry related to the unit circle.

- Instruction includes discussing the definitions of congruent polygons and similar polygons based on corresponding parts. Students should understand that if a problem involves polygons, they will have to use the definitions of congruent and similar (there are no congruence or similarity criteria for polygons) to show the polygons are congruent or similar and use this information to solve the task. (MTR.4.1)
 - When two polygons are congruent, corresponding sides and corresponding angles are congruent.
 - When two polygons are similar, corresponding angles are congruent and corresponding sides are in proportion.
- Problem types includes using congruence and similarity criteria to determine whether two triangles are congruent or similar, and using the definition of congruence and similarity to find missing angle measures and side lengths.
- Instruction includes the understanding of the geometric mean in right triangles and its connection to similarity criteria.

- Geometric Mean Altitude Theorem

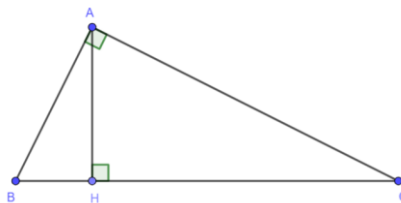
In a right triangle, the altitude from the right angle to the hypotenuse separates the hypotenuse into two segments. The length of the altitude, in triangle ABC , is the geometric mean between the lengths of the two line segments the altitude creates on the hypotenuse. Therefore, by applying the fact that

$\triangle ABC \sim \triangle HBA \sim \triangle HAC$, students can conclude that $\frac{HC}{HA} = \frac{HA}{HB}$, which is equivalent to $HC \cdot HB = (HA)^2$, which is equivalent to $\sqrt{HC \cdot HB} = HA$.



- Geometric Mean Leg Theorem

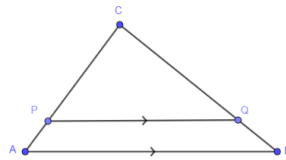
In a right triangle, the altitude from the right angle to the hypotenuse separates the hypotenuse into two segments. The length of one of the legs, in triangle ABC , is the geometric mean between the length of the hypotenuse and the line segment of the hypotenuse adjacent to that leg. Therefore, by applying the fact that $\triangle ABC \sim \triangle HBA \sim \triangle HAC$ students can conclude that $\frac{HB}{BA} = \frac{BA}{BC}$, which is equivalent to $HB \cdot BC = (BA)^2$, which is equivalent to $\sqrt{HB \cdot BC} = BA$.



- Instruction includes the connection between triangle similarity and the Triangle Proportionality Theorem, or Side-Splitter Theorem. Students can explore and conclude that if a line is parallel to one side of a triangle intersecting the other two

sides of the triangle, then the line divides these two sides proportionally.

- For example, given triangle ABC and $\overline{PQ} \parallel \overline{AB}$, students can begin the proof that $\frac{PA}{CP} = \frac{QB}{CQ}$ by first proving $\triangle ABC \sim \triangle PQC$ using the Angle-Angle (AA) criterion. Since corresponding sides of similar triangles are in proportion, students can determine the relationship $\frac{CA}{CP} = \frac{CB}{CQ}$. Students should be able to realize that $CA = CP + PA$ and $CB = CQ + QB$ using the segment addition postulate. Therefore, $\frac{CA}{CP} = \frac{CB}{CQ}$ can be written as $\frac{CP+PA}{CP} = \frac{CQ+QB}{CQ}$. Students can use their algebraic reasoning to rewrite this relationship equivalently as $\frac{CP}{CP} + \frac{PA}{CP} = \frac{CQ}{CQ} + \frac{QB}{CQ}$ which is equivalent to $1 + \frac{PA}{CP} = 1 + \frac{QB}{CQ}$, which is equivalent to $\frac{PA}{CP} = \frac{QB}{CQ}$.



Common Misconceptions or Errors

- Students may have difficulty separating overlapping similar or congruent triangles. To help address this misconception, have students draw the two triangles separately with their corresponding known measures and lengths.
- Students may expect that there are congruence or similarity criteria for non-triangular polygons that are like the criteria for triangles. To help address this misconception, have students create examples in which two quadrilaterals are not congruent even though they have corresponding congruent sides to see that there is no Side-Side-Side-Side congruence criterion for quadrilaterals.

Instructional Tasks

Instructional Task 1 (MTR.7.1)

An artist rendering for the Hapbee Honey Company logo is on a 24" x 36" canvas. The company wants to use the logo on a postcard and is determining the size of the logo based on the different mailing costs. According to the United States Postal Service, mailing costs are determined using the following information.

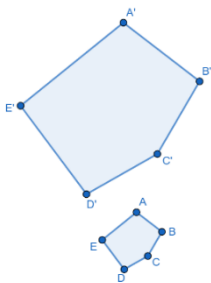
Postcard Size	Price
First-Class Mail® Postcards Maximum size: 6 inches long by 4.25 inches high by 0.016 inch thick	\$0.40
First-Class Mail® Stamped Large Postcards Maximum size: 11.5 inches long by 6.125 inches high by 0.25 inch thick	\$0.58

Part A. What is the maximum length of the postcard if it is similar to the original rendering and falls within the First-Class Mail® Postcards dimensions?

Part B. What is the maximum width of the postcard if it is similar to the original rendering and falls within the First-Class Mail® Stamped Large Postcards dimensions?

Instructional Task 2 (MTR.3.1)

Polygons $ABCDE$ and $A'B'C'D'E'$ are similar and shown.



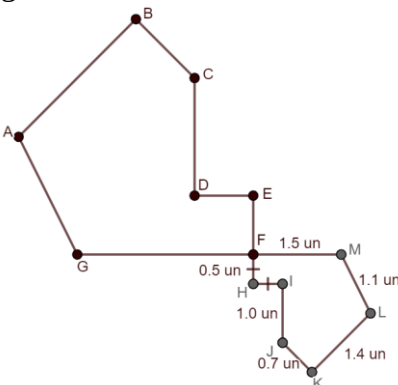
Part A. If $m\angle A' = 103^\circ$, what is the measure of angle A ?

Part B. If $m\angle D = 97^\circ$, what other angle has a measure of 97° ?

Part C. Find the value of x if $DC = 2x + 1.5$, $D'C' = 5.1$ and $\frac{BC}{B'C'} = \frac{1}{3}$.

Instructional Task 3 (MTR.5.1)

Figure $ABCDEFG$ is similar to Figure $LKJIHFM$ with a scale factor of 0.5. Assume that the measure of angle B within Figure $ABCDEFG$ is 90° .



Part A. If point F is located at the origin and line segments EF and CD are vertical and line segments GF and DE are horizontal, determine possible coordinates of each of the points, except points A and B , on Figure $ABCDEFG$.

Part B. What is the perimeter of Figure $ABCDEFG$?

Part C. What is the length of segment DE ?

Part D. What is the perimeter of triangle AGC ?

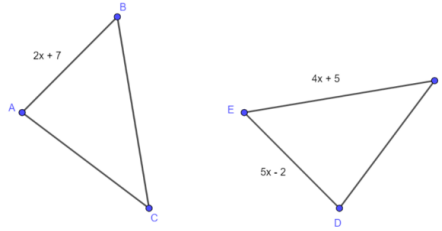
Instructional Items

Instructional Item 1

Triangles ABC and DEF are shown where $\angle A \cong \angle D$, $\angle C \cong \angle F$ and $\overline{AC} \cong \overline{DF}$,

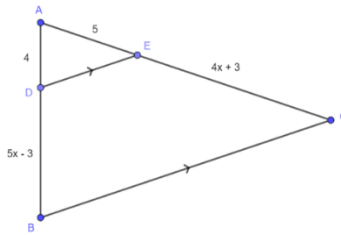
Part A. Determine whether the triangles are congruent.

Part B. If the triangles are congruent, find EF , in units.



Instructional Item 2

If $\triangle ADE$ and $\triangle ABC$ are similar, what is the length of $\overline{AC}$, in units?



**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*